EXACT ENUMERATION / DATA / VERIFICATION

Rank-5 fusion rings

A reproducible census through multiplicity 20

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Abstract

We explain the exact enumeration of all based fusion rings of rank 5 through multiplicity 20, comprising 40,999 isomorphism classes. The account separates the mathematical coverage argument, the completed search, and independent verification of the emitted tensors. All duality types are included, and no categorifiability filter is applied. Code, machine-readable counts, full multiplication tensors and provenance records accompany the text.

AI-generated research · independent verification pending. This manuscript, its code and data were developed with AI assistance. Independent human verification and Lean formalization are pending. Treat the results with caution: exact computational checks establish only their stated scope.

Scope. A census of based rings, not a classification of tensor categories. Only completed whole-rank results are released.

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1 Objects, equivalence and the counting convention

A *fusion ring* here is a free abelian group with a distinguished finite basis $B = \{b_0, b_1, \dots, b_{r-1}\}$, an associative multiplication with unit b_0, and an involution $i \mapsto i^*$ fixing 0. Its structure constants satisfy

$$b_i b_j = \sum_k N_{ij}^k b_k, \qquad N_{ij}^k \in \mathbb{Z}_{\geq 0},$$

$$N_{0j}^k = N_{j0}^k = \delta_{jk}, \qquad N_{ij}^0 = \delta_{i^*,j}, \tag{1}$$

$$N_{ij}^k = N_{i^*k}^j = N_{kj^*}^i. \tag{2}$$

These conventions agree with the usual based-ring definition; see [1, 2]. No assumption of categorifiability, commutativity or integrality of Frobenius–Perron dimensions is imposed unless proved in the relevant rank.

A based isomorphism is a bijection of the distinguished bases preserving the unit and multiplication. In tensor notation it is a permutation q fixing 0 such that

$$N_{ij}^k = N'^{q(k)}_{q(i)q(j)} \quad \text{for every } i, j, k.$$

The coefficient of the unit determines the involution, so such an isomorphism automatically intertwines duality. The *multiplicity* is the maximum entry of the *full* ordered multiplication tensor. It is at least one, including for group rings. Put

$$c_r(m) = \#\{\text{rank-}r \text{ fusion rings of multiplicity exactly } m\}/\cong, \qquad C_r(M) = \sum_{m=1}^{M} c_r(m).$$

A list through M contains all rings of multiplicity at most M, not only those of multiplicity equal to M. This distinction is essential when exporting OEIS b-files.

1.1 Why duality types can be searched separately

An involution on the $r-1$ nonunit elements consists of p disjoint transpositions and $r-1-2p$ fixed points, with $0 \leq p \leq \lfloor (r-1)/2 \rfloor$. Fix one representative d_p of each type. An isomorphism between two tensors with involution d_p commutes with d_p. Hence it is sufficient to quotient that stratum by the centralizer

$$Z_{S_{r-1}}(d_p), \qquad |Z_{S_{r-1}}(d_p)| = 2^p p!(r-1-2p)!.$$

Different p cannot be isomorphic. Therefore a whole-rank census is complete exactly when every required stratum has been searched exhaustively. Finishing only the non-self-dual strata does not certify a whole-rank counting term.

2 The certified release at rank 5

The released bound is $M = 20$ and the cumulative number of classes is $C_5(20) = 40,999$. Every duality type is included. The accompanying full tensor file has one record per based-isomorphism class.

Exact multiplicity m	$c_5(m)$
1	16
2	37
3	82
4	134
5	209
6	336
7	477
8	733
9	863
10	1,082
11	1,383
12	1,948
13	2,211
14	2,554
15	2,936
16	4,157
17	4,255
18	5,080
19	5,640
20	6,866

The data and their checksums are recorded in `results/census.json`. The corresponding completed-run evidence and independent audit are under `verification/rank5/`.

3 An independent census

3.1 Commutativity and coordinates

The following based-ring argument justifies the commutative specialization. Formal-codegree facts used here are recalled in [3].

Lemma 3.1. *Every rank-five fusion ring is commutative.*

Proof. The complex algebra of a Frobenius based ring is semisimple: the coefficient-of-unit functional gives the positive inner product making the basis orthonormal and multiplication a $*$-representation. A noncommutative algebra of dimension five with a one-dimensional representation must be $\mathbb{C} \oplus M_2(\mathbb{C})$. Its unique one-dimensional character is Galois invariant, hence rational, and its values are algebraic integers, hence integers. This character is the Frobenius–Perron dimension character. Its values are positive. Consequently its formal codegree is an integer $D \geq 5$, the sum of five nonzero integer squares. If f denotes the formal codegree of the two-dimensional representation, the standard identity $\sum_E \dim(E)/f_E = 1$ gives

$$f = \frac{2D}{D-1} = 2 + \frac{2}{D-1}.$$

Formal codegrees are algebraic integers. The displayed rational number is not an integer for $D \geq 5$, a contradiction. □

Fix the unit 0 and one of the three dualities

$$(0,1,2,3,4), \quad (0,1,2,4,3), \quad (0,2,1,4,3).$$

Write $C_{ijk} = N_{ij}^{k^*}$. Commutativity and reciprocity say that C is a symmetric tensor, invariant under simultaneous duality. The nonunit triples therefore give respectively $20, 13, 10$ variables, ordered lexicographically by their smallest orbit representative. All variables range independently in $[0, M] \cap \mathbb{Z}$ before associativity is imposed. The unit and duality determine the other entries. Commutativity of the five multiplication matrices is equivalent to associativity: the common unit is cyclic for the span of the matrices, and commuting matrices acting identically on the unit agree everywhere. Direct expansion gives $21, 12, 13$ distinct quadratic equations, respectively. The supplement regenerates these polynomials from the axioms; no source census enters their construction.

3.2 Self-dual rings: reconstruction from a Krylov matrix

In the self-dual case use the ordered triples

$$111, 112, 113, 114, 122, 123, 124, 133, 134, 144,$$
$$222, 223, 224, 233, 234, 244, 333, 334, 344, 444$$

as coordinates $v_0, \dots, v_{19}$. By relabelling, arrange that v_1 is the minimum of the twelve mixed repeated-index coefficients, and $v_1 \le v_2 \le v_3$. The first nontrivial fusion matrix is

$$N_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & a & u^{\mathsf{T}} \\ 0 & u & Q \end{pmatrix}, \qquad u = (v_1, v_2, v_3)^{\mathsf{T}}, \quad Q = \begin{pmatrix} v_4 & v_5 & v_6 \\ v_5 & v_7 & v_8 \\ v_6 & v_8 & v_9 \end{pmatrix}, \quad a = v_0.$$

For each of the nine bounded coordinates of (u, Q) set

$$K = [u, Qu, Q^2u], \quad s = u^{\mathsf{T}}u, \quad t = u^{\mathsf{T}}Qu, \quad H = Q^2 - aQ + uu^{\mathsf{T}} - I.$$

For $i = 1, 2, 3$, let T_i be the bottom-right 3×3 block of N_{i+1}; thus $(T_i)_{jk} = C_{i+1,j+1,k+1}$. The vectors e_i below are the three standard coordinate vectors. Commuting with N_1 gives

$$\begin{aligned} T_i u &= He_i, \\ T_i Qu &= (QH - sQ + u(Qu)^{\mathsf{T}})e_i, \\ T_i Q^2 u &= (Q^2H - sQ^2 - tQ + (Qu)(Qu)^{\mathsf{T}} + u(Q^2u)^{\mathsf{T}})e_i. \end{aligned} \tag{3}$$

These follow successively from the lower blocks of the commutator. If $\det K \neq 0$, multiplication by $\operatorname{adj}(K)$ reconstructs all ten remaining coefficients, affinely in a. One integrality condition is a linear congruence $A + aB \equiv 0 \pmod{\det K}$. A gcd calculation gives every bounded solution for a, including the case $B = 0$. For each such solution all remaining divisibilities, coefficient bounds and all 21 associativity equations are checked. Thus there is no search over the last ten coefficients in this stratum.

3.3 Every singular stratum

The singular cases are essential to completeness. Multiplication by a ring element is zero if it annihilates the unit. Hence any polynomial annihilating the unit under N_1 annihilates N_1 itself. Apply this to the cyclic block generated by the unit.

If $\operatorname{rank} K = 2$, write $Q^2u = \alpha u + \beta Qu$ and $\lambda = \operatorname{tr} Q - \beta$. The invariant rational subspaces of Q show that α, β, λ are integers. The remaining one-dimensional eigenspace of Q has eigenvalue λ. The cyclic block condition is

$$(\lambda^2 - a\lambda - 1)(\lambda^2 - \beta\lambda - \alpha) - \lambda(s\lambda + t - \beta s) = 0. \tag{4}$$

This is linear in a. All zero-coefficient cases are retained.

If $\operatorname{rank} K = 1$, write $Qu = \lambda u$ and let $x^2 - Tx + U$ be the characteristic polynomial on $u^\perp$. The cyclic block polynomial is

$$p(x) = x^3 - (a + \lambda)x^2 + (a\lambda - 1 - s)x + \lambda.$$

For two distinct complementary eigenvalues require $x^2 - Tx + U \mid p$; for a repeated eigenvalue $\eta = T/2$ require $p(\eta) = 0$. These are again linear conditions on a, with no division by a potentially zero coefficient.

For both positive-rank singular cases, the 15 commutator equations linear in the ten remaining coordinates are solved by fraction-free elimination. Enumerate every bounded integral value of the free coordinates, back-substitute, and test all 21 quadratic equations.

If $u = 0$, the unit is annihilated by $N_1^2 - aN_1 - I$. For $a > 0$, $x^2 - ax - 1$ is irreducible over $\mathbb{Q}$, and a five-dimensional rational vector space cannot be a module over the resulting quadratic field. Thus $a = 0$ and $Q^2 = I$. A nonnegative integral symmetric orthogonal matrix is a permutation matrix. The transpositions are handled by the same linear completion. For $Q = I$, the remaining ten variables form the weighted rank-four associativity system: the constant 1 in each diagonal rank-four equation is replaced by 2, because both 1 and the invertible element contribute. The Diophantine elimination of the accompanying rank-four derivation applies with this substitution. All zero branches are implemented explicitly in `weighted.hpp`. For example, its terminal branch $b = c = e = 0$ gives $a = 1$, $d = f = 2$ and $gi + hj = h^2 + i^2 - 6$; it must not be omitted.

3.4 Nonself-dual rings and isomorphisms

For one dual pair, use the 13 orbit coordinates $v_0, \ldots, v_{12}$. Put $A = v_6 - v_5$, $B = v_{10} - v_9$, $C = v_{11} - v_{12}$. Three associativity consequences are

$$Cv_5 = Av_2 + Bv_4, \qquad Cv_9 = Av_4 + Bv_8,$$
$$v_{11}^2 = 1 + v_{12}^2 - v_5^2 + v_6^2 - v_9^2 + v_{10}^2.$$

Enumerate $v_5, v_6, v_9, v_{10}, v_{12}$ and recover v_{11} by an exact square test. Enumerate v_4 and solve for v_2, v_8, retaining all free choices if A or B vanishes. Linear commutators determine or bound the four remaining variables v_0, v_1, v_3, v_7; check all equations. For two dual pairs the corresponding initial reductions are

$$v_0^2 = 1 + v_1^2 - v_2^2 - v_3^2 + 2v_4^2, \qquad v_7^2 = v_2^2 - v_3^2 + v_5^2.$$

Two linear equations determine v_6 unless their coefficients vanish; that branch retains every bounded value. Solve for v_8, v_9 by linear completion and check all 13 equations. The orbit definitions in `census_system2.json` and `census_system4.json` fix these coordinate conventions exactly.

Finally minimize over the permutations preserving the fixed duality: $24, 4, 8$ permutations in the three cases. Distinct duality types cannot be isomorphic. The independent verifier additionally minimizes every output tensor over all 24 unit-fixing permutations.

3.5 The modular prefilter does not change the solution set

In the nonsingular branch each reconstructed coefficient has the form $(A + Ba)/D$. When D is invertible modulo 257, an allowed integer coefficient must have a residue in $[0, M]$. Intersecting precomputed masks for two such coefficients eliminates impossible a values. Every retained value still undergoes exact divisibility and the original integer checks. If $257 \mid D$, the original gcd/congruence calculation is used. All integer-singular branches are unchanged. This is a necessary-condition speedup, not an assumption that a generic determinant is nonzero.

4 What the independent verification establishes

The enumerator and verifier have separate mathematical tasks. Exhaustiveness uses the complete coordinate parameterization, the safe-pruning proof and the normal completion of every required branch. The verifier reads only the emitted *full tensors*; it neither reconstructs the parameterization nor trusts the generator's polynomial list.

For every tensor it checks nonnegative integral coefficients, both unit laws, uniqueness and involutivity of the dual, both reciprocity identities, and every integer associativity identity for all four indices. It then canonicalizes under *every* unit-fixing basis permutation. A matching hash is only a lookup aid: actual canonical tensors are compared. This detects both literal repetitions and different labellings of an isomorphic based ring.

Passing these checks proves soundness and uniqueness of the emitted data; by itself it would not prove that no ring was omitted. That is why normal completion logs and the mathematical coverage argument are retained separately. Comparison with the original ancillary file is an additional containment/regression test, not an input to enumeration and not a replacement for an exhaustiveness proof.

4.1 Provenance and reproducibility

The retained Vercleyen–Slingerland v4 ancillary file has SHA-256

`b587c3f683157369da7cc090f762bbff0e18dcf11acc3f5031d6bc0c643843f8`

Its 28,451 records contain 25,138 distinct classes. The 3,313 redundant records occur at rank five and exact multiplicities 9–12. Their original and corrected counts are respectively

$$(1463, 1794, 2283, 3049) \quad \longrightarrow \quad (863, 1082, 1383, 1948).$$

Every deletion has an explicit basis-permutation witness in the audit folder. This audit does not establish completeness of the source's partial-search cells.

The primary manifest `results/census.json` records every released count, bound, filename and checksum. Compressed tensors use the original line-per-ring, nested-curly-brace format. An entry is $N[i][j][k] = N_{ij}^k$: the output index has already been dualized where the internal parameter convention requires it. The raw tensor needs no additional index conversion.

4.2 Certified datasets and reproducibility

The primary manifest gives the complete bound, exact counts and data checksums for every rank. Each `verification/rankR/` directory records the independent audit and source identity for that dataset. For the laptop censuses it also retains the completed enumeration, commands, environment, phase timings and nonempty subprocess logs. Every required duality stratum completed normally; independent canonicalization found zero duplicate classes.

The independent checks of ranks five through eight ran on GitHub using the supplied tensors. They did not rerun enumeration. The per-rank reports identify the exact verification revision and run, and certify agreement with the established exact-multiplicity count prefix. See `verification/README.md` for the evidence index and `docs/PROVENANCE.md` for attribution.

The local launcher has no clock limit, while the manual hosted workflow uses a finite shared deadline covering every computational phase. A timeout never certifies the unfinished bound. Only a complete, independently verified whole-rank candidate can enter the release; a later failed attempt cannot replace it. Reproduction commands and supported limits are in `docs/REPRODUCIBILITY.md`.

5 Reading and reusing the data

The result files count based rings, not monoidal categories or realizations of a ring. One tensor can admit several inequivalent categorifications, or none. No number here is a classification of fusion categories. The rank-specific count files use exact multiplicity; cumulative totals belong only in the summary column. Proposed OEIS additions must be regenerated from a complete manifest, not copied from an interrupted log.

The retained code was generated with AI assistance and is accompanied by explicit mathematical coverage arguments and independent integer checks. These checks are not a formal proof in a proof assistant. The present text is an explanatory computational companion, not a claim of journal acceptance or external peer review.

6 Reproduction commands

From the repository root on a local computer, request a complete run with no clock limit. An interruption does not certify a smaller census.

```
python3 scripts/run_laptop.py --rank 5 --bound 20
```

To verify the supplied release without recomputing it:

```
python3 scripts/check_release.py --full
```

For a shared-budget frontier search, use `scripts/extend_census.py`. For ranks five through eight it starts at the released bound plus one and restarts the unfinished bound without claiming checkpoint resumption. The local command above verifies all tensors and the old count prefix before packaging a result. The separate timed frontier wrapper bounds every phase. Rank three is instead capped at 1000.

References

[1] P. Etingof, S. Gelaki, D. Nikshych and V. Ostrik, *Tensor Categories*, Mathematical Surveys and Monographs 205, American Mathematical Society, 2015.

[2] G. Vercleyen and J. Slingerland, *On Low Rank Fusion Rings*, Journal of Mathematical Physics 64 (2023), 091703; arXiv:2205.15637v4.

[3] V. Ostrik, *On formal codegrees of fusion categories*, Mathematical Research Letters 16 (2009), 895–901; arXiv:0810.3242.

[4] J. Dong and S. Palcoux, rank-four and rank-five census derivations, project manuscripts and computational supplements, September 2026. The relevant derivations are retained in the repository's `methods/` directory.

[5] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entries A348305, A354471, A354472, A354473, A354475, A354476 and A354477. Proposed updates in this repository are not submitted edits.